

# Sparsity-Adaptive Gated Eviction for Memory-Efficient Long-Context Inference of Large Language Models

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## Declarations

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**Competing interests.** The author has no relevant financial or non-financial interests to disclose.

**Data availability.** The benchmark corpora used in this study (WikiText-2, PG19, LongBench, and the needle-in-a-haystack protocol) are publicly available through their original releases. The complete experimental record that reproduces every table and figure — per-seed values, per-task scores, and raw timing and memory counters — will be released in a public repository upon acceptance.

**Code availability.** The SAGE implementation and all configurations will be released under a permissive license upon acceptance.

**Author contributions.** The sole author performed all parts of the study and wrote the manuscript.

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